DAEN/ISEN 427 In class/homework assignment

Due: Tue Oct 7, 2025

You may use a calculator or short Python snippets. If using Python, include both code and numeric output.

Q1. Over 3 days you observed nearby cars: $x = [3, 4, 1]$. Assume counts are i.i.d. Poisson with rate λ.

a) Compute $\hat{\lambda}_{\text{MLE}}$ and $P(X > 0 \mid \hat{\lambda})$ for a new day. $P(X > 0 \mid \lambda) = 1 - e^{-\lambda}$

Answer: $\hat{\lambda}_{\text{MLE}} = \frac{8}{3} \approx 2.6667$, $P(X > 0 \mid \hat{\lambda}) = 1 - e^{-8/3} \approx 0.9305$.

b) Put a prior $\lambda \sim \text{Gamma}(\alpha = 2, \beta = 2)$ (where β is the *rate*). Compute posterior parameters α', β' for $\lambda \mid x \sim \text{Gamma}(\alpha', \beta')$.

Answer: Posterior: $\alpha' = 2 + 8 = 10$, $\beta' = 2 + 3 = 5$.

c) For a new day X_{new}, use the Gamma–Poisson predictive (Negative Binomial). Show $P(X_{\text{new}} = 0 \mid x) = \left(\frac{\beta'}{\beta'+1}\right)^{\alpha'}$ and compute $P(X_{\text{new}} > 0 \mid x)$.

Answer: Predictive:

$$P(X_{\text{new}} = 0 \mid x) = \left(\frac{5}{6}\right)^{10} \approx 0.1615, \qquad P(X_{\text{new}} > 0 \mid x) = 1 - P(X_{\text{new}} = 0 \mid x) \approx 0.8385.$$

tiny python/colab check:

```python
import math

x = [3,4,1]
lam_mle = sum(x)/len(x)
p_gt0_mle = 1 - math.exp(-lam_mle)
alpha, beta = 2, 2
alpha_p = alpha + sum(x)
beta_p = beta + len(x)
p0_post = (beta_p/(beta_p+1))**alpha_p
p_gt0_post = 1 - p0_post

(lam_mle, p_gt0_mle, (alpha_p, beta_p), p_gt0_post)
```

Q2. A coin has head-probability $p \sim \text{Beta}(\alpha, \beta)$. You flip $N = 10$ times and see $y = 7$ heads.

a) With uniform prior Beta$(1, 1)$, find posterior α', β'.

b) Compute posterior mean $\mu_{\text{post}} = \frac{\alpha'}{\alpha'+\beta'}$.

c) Compute posterior variance

$$\sigma^2_{\text{post}} = \frac{\alpha'\beta'}{(\alpha'+\beta')^2(\alpha'+\beta'+1)}.$$

d) Repeat (a)–(c) with informative prior Beta$(4, 2)$. Compare.

Answer: We are estimating the probability of heads p using a Beta prior and Bernoulli observations. If the prior is

$$p \sim \text{Beta}(\alpha, \beta),$$

and we observe N Bernoulli trials with y heads, then the posterior is

$$p \mid \text{data} \sim \text{Beta}(\alpha', \beta'), \qquad \alpha' = \alpha + y, \quad \beta' = \beta + (N - y).$$

The posterior mean and variance are

$$\mu_{\text{post}} = \frac{\alpha'}{\alpha'+\beta'}, \qquad \sigma^2_{\text{post}} = \frac{\alpha'\beta'}{(\alpha'+\beta')^2(\alpha'+\beta'+1)}.$$

Uniform prior. With Beta$(1, 1)$, $y = 7$ heads out of $N = 10$, we have

$$\alpha' = 1 + 7 = 8, \qquad \beta' = 1 + (10 - 7) = 4.$$

The posterior mean and variance are

$$\mu_{\text{post}} = \frac{8}{8+4} = \frac{8}{12} \approx 0.667, \qquad \sigma^2_{\text{post}} = \frac{8 \cdot 4}{12^2 \cdot 13} = \frac{32}{1872} \approx 0.0171.$$

Informative prior. With prior Beta$(4, 2)$,

$$\alpha' = 4 + 7 = 11, \qquad \beta' = 2 + (10 - 7) = 5.$$

The posterior mean and variance are

$$\mu_{\text{post}} = \frac{11}{11+5} = \frac{11}{16} = 0.6875, \qquad \sigma^2_{\text{post}} = \frac{11 \cdot 5}{16^2 \cdot 17} \approx 0.0126.$$

Interpretation. The informative prior favors heads (since $\alpha = 4,\ \beta = 2$), so the posterior mean is slightly higher (0.6875 vs. 0.667). It also adds extra "pseudo-observations," reducing the posterior variance (0.0126 vs. 0.0171). This demonstrates how the prior affects both the center and the spread of the posterior distribution.

Q3. For a Beta$(1,1)$ prior on p, the posterior mean after N observations is

$$\mu_{\text{post},N} = \frac{y+1}{N+2}.$$

a) Show that after observing $x_{N+1} \in \{0,1\}$,

$$\mu_{\text{post},N+1} = \mu_{\text{post},N} + \frac{1}{N+3}\big(x_{N+1} - \mu_{\text{post},N}\big).$$

b) Starting from no data, process $x = [1,0,1,1,0]$. Compute μ_{post} after each.

Answer: For a Beta$(1,1)$ prior, after N Bernoulli observations with y successes, the posterior is

$$p \mid x_{1:N} \sim \text{Beta}(y+1, N-y+1),$$

and the posterior mean is

$$\mu_{\text{post},N} = \frac{y+1}{N+2}.$$

Suppose we observe a new data point $x_{N+1} \in \{0,1\}$. The posterior after this new point has parameters $y' = y + x_{N+1}$ and $N' = N+1$, giving

$$\mu_{\text{post},N+1} = \frac{y + x_{N+1} + 1}{N+3}.$$

Since $\mu_{\text{post},N} = \frac{y+1}{N+2}$, we can rewrite

$$\mu_{\text{post},N+1} = \frac{(N+2)\mu_{\text{post},N} + x_{N+1}}{N+3} = \mu_{\text{post},N} + \frac{1}{N+3}\big(x_{N+1} - \mu_{\text{post},N}\big).$$

This is the same form as the *Rescorla–Wagner* learning rule:

$$\text{new estimate} = \text{old estimate} + \text{learning rate} \times \text{prediction error}.$$

Here, the learning rate is $\frac{1}{N+3}$, which decreases as more data are observed, and the prediction error is $x_{N+1} - \mu_{\text{post},N}$.

For the sequence $x = [1,0,1,1,0]$, starting from $\mu_0 = 0.5$, we update step by step:

Step	x_N	μ_N
1	1	0.667
2	0	0.500
3	1	0.600
4	1	0.667
5	0	0.571

Example for the first two steps:

$$\mu_1 = 0.5 + \frac{1}{3}(1-0.5) = 0.667, \qquad \mu_2 = 0.667 + \frac{1}{4}(0-0.667) = 0.500.$$

As N grows, the learning rate shrinks, making the posterior mean less sensitive to individual new data points. This illustrates how Bayesian updating with a Beta prior corresponds exactly to incremental learning with decreasing step sizes.

Q4. With flat prior on s, two independent measurements: $x_1 \sim \mathcal{N}(s, \sigma_1^2), \qquad x_2 \sim \mathcal{N}(s, \sigma_2^2)$.

a) Derive or state:

$$\mu_{\text{post}} = \frac{\frac{x_1}{\sigma_1^2} + \frac{x_2}{\sigma_2^2}}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}, \qquad \sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}.$$

b) Numerical practice: $x_1 = 10, \sigma_1 = 2$; $x_2 = 14, \sigma_2 = 4$.

Answer: We have two independent noisy measurements

$$x_1 \sim \mathcal{N}(s, \sigma_1^2), \qquad x_2 \sim \mathcal{N}(s, \sigma_2^2),$$

with a flat (uninformative) prior on s. In this case, the posterior distribution for s is Gaussian, with

$$\mu_{\text{post}} = \frac{\frac{x_1}{\sigma_1^2} + \frac{x_2}{\sigma_2^2}}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}, \qquad \sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}.$$

Each measurement contributes proportionally to its *precision* $w_i = 1/\sigma_i^2$: more precise measurements (smaller σ_i) receive higher weight.

For the numerical example

$$x_1 = 10,\ \sigma_1 = 2; \qquad x_2 = 14,\ \sigma_2 = 4,$$

the precisions are

$$w_1 = \frac{1}{4} = 0.25, \qquad w_2 = \frac{1}{16} = 0.0625.$$

The posterior mean is

$$\mu_{\text{post}} = \frac{0.25 \times 10 + 0.0625 \times 14}{0.25 + 0.0625} = \frac{3.375}{0.3125} = 10.8,$$

which lies closer to $x_1 = 10$ because x_1 is more reliable.

The posterior variance is

$$\sigma_{\text{post}}^2 = \frac{1}{0.25 + 0.0625} = \frac{1}{0.3125} = 3.2, \qquad \sigma_{\text{post}} = \sqrt{3.2} \approx 1.79,$$

which is smaller than either individual variance. Combining measurements increases precision.

In short: the posterior estimate is a precision-weighted average, and the combined uncertainty is lower than that of either single measurement. The more precise cue dominates the estimate.